

Final Technical Summary

Architecture

The system takes an advisor query, retrieves similar real postings from the 80,000-row training corpus, applies a transparent role-fit rubric, and generates a grounded recommendation with cited posting evidence. It does not infer authorization eligibility from missing source text.

Evaluation

System	Rows	Accuracy	Macro F1
Milestone 2 TF-IDF baseline	20,000	79.3%	78.5%
Milestone 3 true PEFT LoRA	4,000	87.5%	87.3%
Final end-to-end system	4,000	80.0%	80.1%

The true PEFT LoRA model is the strongest pure classifier. The final prototype uses retrieval plus a transparent decision layer because the product requirement is a grounded advisor workflow, not only a label.

Risk and grounding

The largest risk is hallucinated work authorization. The generator is deterministic and cites only retrieved fields. Every output warns that CPT, OPT, and sponsorship evidence must be checked separately. The next production step is to evaluate on human-reviewed Career Center postings with real authorization language.